

3-MONTH EURIBOR FUTURES (FEI)

Spread & Curve Structure — Fundamental Reference

The Eurozone STIR Market — ECB Policy, a Surviving Term Rate, and the EURIBOR/ESTR Dual Structure

This document presents a fact-based reference on the fundamental drivers of Three-Month Euribor futures (ticker: FEI on ICE Futures Europe; also traded as FEU3 on Eurex), with particular focus on the forces that shape the forward rate curve and the spread structure between contract months. Euribor futures are the primary exchange-traded instrument for hedging and speculating on the path of euro-area short-term interest rates, and occupy a structurally distinct position among the major STIR futures markets covered in this series: unlike GBP LIBOR and USD LIBOR, which were permanently discontinued and replaced by SONIA and SOFR respectively (covered in companion reference documents), Euribor itself was reformed **but not discontinued**. It remains a live, panel-bank-submitted, forward-looking term rate, administered under a reformed methodology, and continues to underpin both the Euribor futures contract covered in this document and the large majority of euro-denominated floating-rate loans and mortgages. The Eurozone's risk-free overnight rate, €STR (Euro Short-Term Rate), exists alongside Euribor as a separate, parallel benchmark rather than as its replacement, with its own dedicated futures contract — a genuinely distinct market structure from the single-benchmark transition completed in both the US and UK markets. Understanding this dual-benchmark structure, ECB Governing Council mechanics, and the specific macroeconomic and political-economy considerations particular to a diverse 20-member-state currency union is the subject of this document.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Settlement Mechanics

Three-Month Euribor futures are listed on both ICE Futures Europe (ticker FEI) and Eurex (ticker FEU3), the two competing venues for euro-area STIR futures — a dual-listing structure with no precise equivalent among the single-venue SOFR (CME) and SONIA (ICE Futures Europe) markets covered in companion reference documents in this series.

Parameter	Detail
Exchange	ICE Futures Europe (ticker FEI) and Eurex (ticker FEU3) — both venues list economically equivalent contracts on the same underlying rate, a competitive dual-listing structure distinct from the single-venue SOFR and SONIA markets
Underlying	EURIBOR (Euro Interbank Offered Rate) for three-month euro term deposits, administered by the European Money Markets Institute (EMMI)
Contract size	EUR 1,000,000 notional (ICE FEI); Eurex FEU3 is quoted with a EUR 2,500 per index point convention — both structured to deliver an equivalent basis point value for hedging purposes
Quotation	100 minus the annualised 3-month EURIBOR rate. Example: a price of 96.85 implies a rate of 3.15%. The same inverted quotation convention used throughout the global STIR futures market (SOFR, SONIA, and Euribor all share this convention).
Minimum tick	0.005 (half a basis point); on Eurex, the minimum price change is 0.00125 index points (EUR 3.125) for the relevant instrument types — tick conventions differ slightly between the two venues
Contract months	March (H), June (M), September (U), December (Z) — quarterly cycle. Eurex lists the six nearest successive calendar months plus 22 further quarterly months (March/June/September/December cycle), extending up to 72 months (6 years) forward.
Settlement type	Cash-settled. Crucially, this settles to a single point-in-time EURIBOR fixing on the last trading day, NOT a compounded rate over an accrual period — a fundamental structural difference from SOFR and SONIA futures, addressed in full in Section 1.1.
Last trading day	Two exchange days prior to the third Wednesday of the maturity month (Eurex convention); trading in the expiring month ceases at 11:00 CET, with the EMMI Euribor reference rate determination required to occur that day.
Final settlement	Established at 11:00 CET on the final settlement day, based on the EMMI-published EURIBOR reference rate for three-month euro deposits, rounded to three decimal places.
Matching algorithm	Gradual time-based pro-rata (GTBPR), the same general matching convention described for ICE SONIA futures in the companion UK reference document.

1.1 The Defining Structural Feature: A Forward-Looking Term Rate That Survived

This is the single most important conceptual starting point for analysing Euribor futures relative to every other STIR contract covered in this document series, and merits sustained emphasis:

- **Euribor was reformed, not replaced:** following the global benchmark reform process triggered by the LIBOR manipulation scandal, EU and UK regulators required benchmark administrators to either reform their methodology to comply with the EU Benchmarks Regulation (BMR) or face

discontinuation. GBP LIBOR and USD LIBOR were discontinued and replaced by SONIA and SOFR respectively (described in the companion reference documents). EURIBOR, administered by EMMI, instead adopted a reformed, BMR-compliant "hybrid methodology" — combining actual transaction data where available with expert-judgement-based panel bank submissions where transaction volume is insufficient — and continued in continuous operation throughout.

- **Forward-looking term rate, not backward-looking compounded overnight rate:** this is the critical mechanical consequence. SOFR and SONIA futures, as described in the companion SR3 and SO3 reference documents, settle against a **compounded daily rate over the contract's 3-month accrual period** — the final settlement value is only known once that period has concluded. EURIBOR futures, by contrast, settle against a **single, forward-looking term rate fixed on a single day** (the last trading day) — the same general settlement mechanism used by the legacy Short Sterling and Eurodollar contracts before the SOFR/SONIA transition. This means Euribor futures do not require the "days in quarter" FOMC/MPC-meeting-attribution arithmetic described in detail in the SR3 and SO3 reference documents, since the settlement rate is not built up from a sequence of daily overnight fixings across the accrual period.
- **Practical implication for spread and fly trading:** an FEI calendar spread is a more direct expression of the implied forward EURIBOR curve between two specific future dates, without the additional layer of "how many days of the new vs old rate fall within this particular quarter" arithmetic that characterises SR3 and SO3 spread interpretation — a genuinely simpler mechanical structure, though one that still requires careful attention to where ECB Governing Council meeting dates fall relative to each contract's fixing date (Section 2.2).

1.2 Why Euribor Continues to Coexist With €STR

- Unlike the US and UK, where the legacy IBOR was discontinued and a single new risk-free overnight rate (SOFR, SONIA) took over as the dominant benchmark, the Eurozone operates a **dual-benchmark structure**: EURIBOR continues as the dominant reference rate for retail and commercial floating-rate lending (euro-area mortgages and corporate loans overwhelmingly reference Euribor), while €STR (Euro Short-Term Rate), administered by the European Central Bank itself, serves as the Eurozone's risk-free overnight rate benchmark, used predominantly in derivatives markets, particularly for discounting and as the reference for overnight index swaps (OIS).
- **€STR futures as a separate, parallel contract:** CME Group lists Euro Short-Term Rate (€STR) futures as part of its "European Overnight Index futures" suite, structurally and conceptually analogous to SOFR futures (a compounded overnight rate, in this case the Eurozone's risk-free benchmark) but trading as an entirely separate instrument from Euribor futures, not as its replacement. Basis spread futures between €STR and Euribor are also listed, allowing direct trading of the EURIBOR-€STR spread itself.

2. European Central Bank Policy — The Primary Driver

The European Central Bank's Governing Council sets the Eurozone's key policy rates, which anchor euro-area short-term interest rates including both EURIBOR and €STR. Understanding ECB mechanics — and the specific complexity introduced by governing a single monetary policy across twenty distinct member states — is the prerequisite for any analytical framework applied to the FEI curve.

2.1 The ECB's Three Key Policy Rates

- **Deposit Facility Rate (DFR):** the rate paid on commercial bank reserves held overnight at the ECB. Since the ECB restructured its operational framework, the DFR has become the effective primary policy rate and the closest anchor for €STR and short-dated EURIBOR — structurally analogous in function to the Fed's IORB and the Bank of England's Bank Rate described in the companion reference documents, though the ECB maintains three distinct rates rather than the Fed's single target range or the BoE's single Bank Rate.
- **Main Refinancing Operations (MRO) rate:** the rate at which banks can borrow from the ECB for one week against collateral — historically the ECB's primary policy rate before the operational framework shift toward the DFR as the binding rate in a system with abundant excess reserves.
- **Marginal Lending Facility (MLF) rate:** the rate for overnight borrowing from the ECB, typically set as a ceiling above the MRO rate, completing the ECB's "corridor" system of three rates bracketing the overnight market rate.
- **EURIBOR's relationship to the DFR:** because EURIBOR is an unsecured interbank term rate (panel banks submitting rates at which they could borrow unsecured euro funds for three months), it trades at a spread above the DFR reflecting term premium, credit/liquidity considerations among panel banks, and market expectations for the policy rate path over the three-month term — a structurally different relationship from SONIA's close tracking of UK Bank Rate (since SONIA is overnight rather than term) and closer in spirit to how a term rate generally trades versus an overnight policy anchor.

2.2 Governing Council Meeting Schedule and Its Impact on the FEI Curve

The ECB Governing Council holds monetary policy meetings eight times per year (the same eight-meetings-per-year cadence as the Fed's FOMC and the Bank of England's MPC), though the specific mechanical relationship to the FEI curve differs from SR3/SO3 given the settlement mechanism described in Section 1.1:

- Because FEI futures settle to a single forward-looking term fixing rather than a compounded accrual-period rate, the precise date of a Governing Council decision relative to a contract's last trading day matters somewhat differently than the "days in quarter" arithmetic central to SR3/SO3 analysis: an FEI contract maturing shortly after a Governing Council decision will have that decision fully reflected in its settlement EURIBOR fixing, while a contract maturing shortly before a decision will not, with no partial-period blending of the kind that characterises a SOFR/SONIA accrual period spanning a meeting.
- **ECB press conference and the "Lagarde press conference" effect:** the Governing Council decision is followed by a press conference (currently led by President Christine Lagarde) roughly 45 minutes after the rate announcement, during which prepared remarks and the Q&A session frequently move the FEI curve more than the bare rate decision itself — the same general "press conference moves more than the statement" pattern documented for the Fed and Bank of England in the companion reference documents.

2.3 ECB Staff Macroeconomic Projections

- Published quarterly (March, June, September, December), the ECB staff (and, for the June and December rounds, Eurosystem staff, incorporating national central bank input) macroeconomic

projections provide GDP growth and HICP inflation forecasts presented as point estimates with a range, a different communication format from both the Fed's dot plot (individual rate-path point estimates) and the Bank of England's fan chart (a probability distribution conditioned on market-implied rates) described in the companion reference documents. Notably, the ECB's published projections do not include an explicit policy-rate-path forecast component analogous to either the Fed's dots or the conditioning assumption embedded in the BoE's fan chart, making the FEI curve's relationship to the ECB's own forecasts somewhat less direct than the equivalent SR3/dot-plot or SO3/fan-chart relationship.

3. Macroeconomic Fundamentals — The Rate-Setting Inputs

The ECB's primary mandate is price stability, defined as 2% HICP (Harmonised Index of Consumer Prices) inflation over the medium term — a single primary target broadly similar in structure to the Bank of England's remit, though the practical challenge of assessing "Eurozone inflation" is compounded by the currency union's genuine economic diversity across member states.

3.1 Inflation Data

Indicator	Publisher	Frequency	Release Timing	ECB Target / Reference	Curve Impact
HICP (Eurozone aggregate, flash + final)	Eurostat	Monthly (flash ~end of month; final ~mid-following month)	~11:00 CET	2% over the medium term — the ECB's sole primary mandate reference	The single most important data release for FEI; the flash estimate moves the curve immediately, with the final figure typically confirming rather than surprising
Core HICP (ex food, energy, alcohol, tobacco)	Eurostat	Monthly	Same release as headline	Closely watched underlying inflation signal	Core surprises move the whole FEI curve; a persistent core/headline divergence is closely tracked by the Governing Council
National HICP releases (Germany, France, Italy, Spain)	National statistical offices	Monthly, typically days ahead of the Eurozone aggregate	Staggered through the month	Leading indicators for the Eurozone aggregate, given each country's weight in the HICP basket	German and French prints in particular are watched as early reads ahead of the full Eurozone flash figure, given their large basket weights
Negotiated wage growth tracker	ECB (compiled from national sources)	Quarterly	Variable	An ECB-developed metric specifically created to track Eurozone wage pressure ahead of slower official wage data	Referenced explicitly in ECB communication as a forward-looking input on services-inflation persistence, given the documented lag in standard Eurozone wage statistics
5y5y Inflation Swap Forward (market-implied)	Swap market	Daily	Continuous	Market's long-run Eurozone inflation expectation; ECB monitors for "de-anchoring"	A persistent deviation from target in this market-implied measure is a documented Governing Council communication trigger

3.2 The Core Challenge: A Single Rate for Twenty Diverse Economies

- **Inflation and growth divergence across member states:** unlike the Fed and the Bank of England, which each set policy for a single national economy, the ECB sets a single policy rate for a currency union spanning twenty member states with materially different inflation rates, growth trajectories, labour market structures, and fiscal positions at any given time. A Governing Council decision that is appropriately calibrated for the Eurozone aggregate may be too tight for

slower-growing, lower-inflation member states and too loose for faster-growing, higher-inflation ones — a structural tension with no parallel in the single-economy frameworks described in the companion SR3 and SO3 reference documents, and one that is a recurring, explicit subject of Governing Council and market commentary.

- **Sovereign spread monitoring:** because individual member states issue their own sovereign debt (rather than a single unified Eurozone bond, notwithstanding the post-2020 growth in joint EU issuance), the spread between, for example, Italian BTP yields and German Bund yields (the "BTP-Bund spread") is a closely watched market signal of sovereign stress within the currency union, with sustained widening historically associated with euro-area financial fragmentation risk that has, at various points (most notably the 2010–2012 European sovereign debt crisis), directly constrained the ECB's ability to tighten policy purely on the basis of aggregate Eurozone inflation data.
- **The Transmission Protection Instrument (TPI):** a tool announced by the ECB in 2022, designed to allow the ECB to purchase the sovereign debt of specific member states experiencing "unwarranted" spread widening that threatens monetary policy transmission, intended to give the Governing Council greater latitude to set policy for the aggregate Eurozone without being constrained by fragmentation risk in any single member state — a structurally unique policy instrument with no equivalent in the single-country US or UK frameworks.

3.3 Labour Market and Activity Data

- **Eurozone unemployment rate:** published monthly by Eurostat, an aggregate figure that, consistent with the theme of Section 3.2, masks very substantial variation across member states (historically ranging from low single digits in some Northern European states to double digits in parts of Southern Europe).
- **S&P; Global Eurozone Composite PMI:** monthly, the Eurozone's equivalent to the US ISM and UK CIPS surveys described in the companion reference documents, published alongside individual national PMIs (Germany and France particularly closely watched given their economic weight) that allow market participants to assess intra-Eurozone divergence directly.
- **German Ifo Business Climate Index and ZEW Economic Sentiment:** Germany-specific business and analyst sentiment surveys, watched closely given Germany's position as the Eurozone's largest economy and its particular exposure to global manufacturing and export-cycle dynamics, including the country's well-documented industrial/manufacturing sector softness in recent years.
- **Eurozone GDP (flash + final estimates):** quarterly, Eurostat, with national breakdowns published alongside the aggregate figure allowing direct observation of which member states are driving the aggregate growth (or contraction) figure.

4. The Rate Cycle and FEI Curve Regimes

As with SR3 and SO3, the FEI curve takes on different shapes in different phases of the monetary policy cycle. The general regime framework described in the companion reference documents applies conceptually to FEI; this section highlights the Eurozone-specific considerations relevant to identifying and trading each regime.

Regime	Rate Direction	Curve Shape	Eurozone-Specific Consideration
Hiking cycle	Rising rates	Inverted (front > back in price). Curve flattens as hikes priced in.	Eurozone hiking cycles carry an explicit fragmentation-risk monitoring requirement (Section 3.2) — the pace of hikes has, historically, been calibrated with some sensitivity to peripheral sovereign spread behaviour, a consideration absent from the single-economy Fed and BoE frameworks.
Peak / pause	On hold at terminal rate	Flat to mildly inverted near front; back curve may begin pricing cuts.	Divergent member-state conditions at the pause stage can produce more vocal public disagreement among Governing Council members (who include the national central bank governors of all member states) than is typical of FOMC or MPC commentary, given the genuine diversity of conditions each governor represents.
Cutting cycle	Falling rates	Front falls in yield (rises in price); back anchored by long-run neutral rate expectations.	Eurozone cutting cycles are watched for their differential effect across member states with different mortgage market structures — some (e.g., parts of Southern Europe) have historically had a higher share of floating-rate, EURIBOR-linked mortgages than others (e.g., France, with a larger fixed-rate mortgage market share), meaning the same ECB rate move passes through to household finances at different speeds across the currency union.
Post-cycle / normalisation	Approaching neutral	Flat or mildly upward sloping in yield terms.	Eurozone neutral rate (r^*) estimates are subject to the same general estimation uncertainty described for the US and UK, with the added complexity that a single neutral rate concept is being applied across structurally different national economies.

4.1 The Neutral Rate and the Term Premium

- As with the Fed and Bank of England, the ECB does not publish an official Eurozone neutral rate (r^*) estimate, though ECB staff research and Governing Council member speeches periodically reference internal model-based estimates, subject to the same general estimation uncertainty described in the companion SR3 reference document's discussion of the Holston-Laubach-Williams framework, here further complicated by the question of whether a single Eurozone-wide r^* concept is even well-defined across economically diverse member states.
- German Bund yields as the Eurozone's "risk-free" anchor:** because German Bunds are conventionally treated as the Eurozone's benchmark risk-free sovereign curve (analogous to the role US Treasuries and UK gilts play in their respective markets), Bund yield movements — and the German fiscal policy developments that affect Bund supply, including Germany's historically conservative "debt brake" fiscal framework and any changes to it — are a particularly important

input to the FEI back curve's term premium, distinct from the BTP-Bund spread monitoring described in Section 3.2, which concerns peripheral fragmentation risk specifically.

5. Key Data Events and the FEI Market Calendar

The FEI curve responds to a well-defined calendar of Eurozone data releases and ECB policy events, structurally parallel to the US and UK calendars described in the companion reference documents, with the added complexity of staggered national data releases ahead of Eurozone aggregate figures.

Event / Release	Publisher	Frequency	Timing	Primary Curve Impact	Magnitude
HICP Flash Estimate	Eurostat	Monthly	Last business day of reference month, ~11:00 CET	All curve — the single largest monthly event	★★★★★
ECB Governing Council Decision	ECB	8× per year	14:15 CET (decision); 14:45 CET (press conference)	All curve — immediate repricing, with the press conference often moving more than the bare decision	★★★★★
ECB Staff/Eurosysteem Macroeconomic Projections	ECB	4× per year, with Governing Council decision	Published with the March/June/September/December decisions	C2-C12 — growth/inflation forecast revision	★★★★★
National flash HICP (Germany, France, Spain, Italy)	National statistical offices	Monthly, staggered ahead of Eurozone flash	Variable, days before Eurozone release	C1-C4 — early read on the Eurozone aggregate	★★★
S&P; Global Eurozone Composite PMI	S&P; Global	Monthly	~3rd-5th business day	C1-C4 — activity leading indicator, watched alongside national breakdowns	★★★
German Ifo Business Climate	Ifo Institute	Monthly	~Last week of month	C1-C4 — Germany-specific manufacturing/export sentiment signal	★★
Eurozone GDP (flash + final)	Eurostat	Quarterly	~Mid-month following quarter end	C2-C8 — growth trajectory, with national breakdowns	★★★
ECB Negotiated Wage Tracker	ECB	Quarterly	Variable	C2-C8 — forward-looking wage pressure signal explicitly cited by the Governing Council	★★★
ECB Account of the Monetary Policy Meeting ("Minutes")	ECB	8× per year, ~4-week lag	Variable	C3-C8 — debate nuance; published with a longer lag than the BoE's same-day minutes	★★★
BTP-Bund and other peripheral sovereign spreads	Market, continuous	Continuous	Market hours	All curve, acutely during stress episodes — fragmentation-risk signal unique to the currency-union structure	★★★

Bank Lending Survey (BLS)	ECB	Quarterly	~3 weeks before Governing Council meeting	C2-C8 — credit conditions and loan demand, an explicit Governing Council input on transmission strength	★★
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5.1 The HICP Flash Estimate — Anatomy of the Most Important Event

- The Eurozone HICP flash estimate, released on the last business day of the reference month (a faster turnaround than the equivalent US and UK CPI release timing), is consistently the single most market-moving scheduled event for the FEI curve.
- **The staggered-national-release dynamic:** because individual member states (Germany, France, Spain, and others) publish their own national flash HICP figures in the days immediately preceding the Eurozone aggregate release, the market frequently has a reasonably well-informed expectation for the Eurozone figure before it is published, given each country's known weight in the HICP basket — a structural feature of the Eurozone data calendar with no direct parallel in the single-release US CPI or UK CPI calendars described in the companion reference documents, and one that can moderate (though by no means eliminate) the surprise-driven volatility around the Eurozone aggregate release itself.
- **Core HICP and the energy/food volatility consideration:** as with the US and UK frameworks, core HICP (excluding the more volatile food and energy components) is watched as the underlying inflation signal, with the Eurozone's historically higher direct energy-import dependence (particularly pre-2022 reliance on Russian pipeline gas, addressed further in Section 8) having made the headline-core HICP gap a particularly large and closely tracked figure during the 2022 European energy price shock.

6. Spread and Fly Mechanics on the FEI Curve

6.1 What a Spread Expresses — Without the Accrual-Period Complication

Because FEI futures settle to a single forward-looking fixing rather than a compounded accrual-period rate (Section 1.1), an FEI calendar spread is a more direct expression of the implied forward EURIBOR curve between two specific future dates than the equivalent SR3 or SO3 spread:

- **Positive spread (front > back in price):** the market implies the 3-month EURIBOR rate fixed at the back contract's date will be lower than the rate fixed at the front contract's date — i.e., the market expects rates to fall between the two dates.
- **Negative spread:** the market expects the 3-month EURIBOR fixing to rise between the two dates.
- **No "days in quarter" attribution required:** because each contract settles to a single point-in-time fixing rather than a compounded average over its own reference period, there is no need to assess what fraction of an accrual period a given ECB decision affects — each contract's settlement value simply reflects market expectations for the 3-month EURIBOR rate as of that specific future date, a genuinely simpler mechanical structure than the SR3/SO3 equivalent described in the companion documents.

6.2 What a Fly Expresses

- An FEI fly (long the wings, short twice the body) captures curvature in the implied forward EURIBOR curve — the market's view of whether the rate path between two dates is expected to move smoothly or to contain a discrete kink, most commonly used to express a view on whether a specific Governing Council meeting (or sequence of meetings) falling between the wing contracts' fixing dates will deliver an outcome different from what a smooth interpolation between the wings would imply.
- **Eurozone-specific fly consideration:** because Governing Council meetings are accompanied by ECB staff macroeconomic projections only four times per year (Section 2.3), flies spanning a "projections meeting" can behave differently from flies spanning a non-projections meeting, structurally analogous to the Report-meeting/non-Report-meeting asymmetry described for the Bank of England's SO3 market in the companion document.

6.3 Curve Zone Characteristics

Zone	Contracts (approx.)	Dominant Driver	Key Signals	Volatility
Front	C1-C4 (~3-12 months)	Near-term Governing Council path; HICP flash and national pre-releases dominate	Monthly HICP flash, national HICP pre-releases, Lagarde press conference language	Highest
Belly	C4-C8 (~12-24 months)	Terminal rate expectations; ECB staff projections; Eurozone growth trajectory	Quarterly ECB projections, Eurozone Composite PMI, German Ifo	High
Mid	C8-C20 (~24-60 months)	Neutral rate (r^*) estimates; Bund term premium; fragmentation-risk background	BTP-Bund spread trend, German fiscal policy news, ECB staff r^* research	Moderate
Back	C20+ (~60-72 months)	Long-run r^* ; EU-level fiscal integration trajectory; structural Eurozone growth debate	EU joint-issuance developments, long-run Eurozone competitiveness commentary	Lowest

7. ECB Balance Sheet — APP, PEPP, and the TLTRO Legacy

As with the Federal Reserve and Bank of England balance sheets described in the companion reference documents, the ECB's balance sheet — expanded through several distinct programmes since 2014–15 and now contracting — affects the FEI curve through reserve supply and term premium channels, with several Eurozone-specific programme features.

7.1 Asset Purchase Programme (APP) and Pandemic Emergency Purchase Programme (PEPP)

- **APP:** the ECB's primary QE vehicle, comprising several sub-programmes targeting public sector bonds, corporate bonds, covered bonds, and asset-backed securities — broadly analogous in function to the Fed's QE and the Bank of England's APF described in the companion documents, though structured across multiple distinct sub-programmes targeting different asset classes.
- **PEPP:** a separate, larger emergency programme launched in response to the COVID-19 pandemic, notable for its explicit flexibility to deviate from the ECB's normal "capital key" purchase allocation (which otherwise allocates purchases across member states in proportion to their ECB capital contribution), giving the ECB greater latitude to concentrate purchases in specific member states experiencing acute stress — a flexible fragmentation-management design feature with no precise equivalent in the single-country Fed or BoE QE programmes.

7.2 Targeted Longer-Term Refinancing Operations (TLTRO)

- A distinctly Eurozone policy tool with no direct Fed or BoE parallel: TLTROs offer Eurozone banks long-term funding from the ECB at favourable rates, conditional on banks maintaining or expanding lending to the real economy — a targeted credit-easing instrument layered alongside the ECB's asset purchase programmes. The wind-down and repayment of outstanding TLTRO balances has, at various points, been a relevant background consideration for euro-area bank funding costs and, indirectly, EURIBOR panel bank submission behaviour.

7.3 Quantitative Tightening — Differentiated Pace Across Programmes

- The ECB's QT process has, at various points, proceeded at a differentiated pace across the APP and PEPP portfolios specifically — a more granular, programme-by-programme QT approach than the Fed's single passive-runoff framework or the Bank of England's combined passive-runoff-plus-active-sales approach described in the companion documents, reflecting the ECB's judgement about the relative importance of reinvestment flexibility (particularly under PEPP) for managing fragmentation risk.

8. Cross-Market Relationships

8.1 FEI and the German Bund / Peripheral Sovereign Curve

- As described in Section 4.1, German Bund yields are the Eurozone's conventional risk-free sovereign benchmark, while peripheral sovereign yields (Italy, Spain, Portugal, Greece) and their spread to Bunds carry the fragmentation-risk information described in Section 3.2 — a genuinely two-dimensional sovereign curve structure (a risk-free benchmark plus a fragmentation-risk spread) that has no equivalent in the single-sovereign-curve US Treasury or UK gilt markets described in the companion reference documents.

8.2 FEI and the Euro (EUR)

- Higher Eurozone short rates (lower FEI prices) typically support EUR by increasing EUR-denominated asset returns relative to foreign alternatives, the same general mechanism described for SR3/USD and SO3/GBP in the companion documents.
- **EUR/USD as the most heavily traded cross-currency pair globally:** given EUR/USD's position as the single most liquid currency pair in global FX markets, the Fed-ECB rate differential carries a degree of global market attention somewhat beyond the equivalent Fed-BoE or ECB-BoE relationships, with EUR/USD moves themselves feeding back into US and global financial conditions in a manner the ECB and Fed both monitor.

8.3 FEI and Eurozone Banking Sector Health

- Because EURIBOR remains a panel-bank-submitted rate (Section 1.1), the perceived credit and liquidity health of the contributing panel banks is, in principle, a more direct input into the EURIBOR fixing itself than is the case for SOFR or SONIA, which are calculated from broad, deep transaction volumes rather than a smaller panel's own funding-cost assessment. Episodes of Eurozone banking sector stress (historically including periods during the European sovereign debt crisis) have, at various points, widened EURIBOR's spread to €STR as a reflection of elevated interbank credit/liquidity premia.

9. Global Context — the Fed-ECB-BoE Triangle and Energy Dependence

9.1 The Fed-ECB-BoE Relative Timing Relationship

- As described from the UK perspective in the companion SO3 reference document, ECB and Fed policy cycles interact through EUR/USD and global financial conditions channels, with the relative timing of hikes and cuts across the three major Western central banks (Fed, ECB, BoE) a recurring subject of comparative market commentary, though — as noted in the SO3 document — this is an empirical pattern rather than a structural rule.

9.2 European Energy Dependence — A Structural Eurozone-Specific Factor

- The Eurozone's historically substantial dependence on imported energy, particularly natural gas (with pre-2022 reliance on Russian pipeline gas a widely documented structural feature of the European energy market), gives European inflation a documented higher sensitivity to global energy price shocks than is generally the case for the more energy-self-sufficient US economy — a structural factor behind the scale of the 2022 European inflation shock and the ECB's subsequent hiking cycle, and a continuing background consideration for assessing the relative inflation sensitivity of the Eurozone economy to any future global energy price disruption.

9.3 EU Fiscal Architecture and Joint Issuance

- Unlike the single sovereign-issuer frameworks of the US (Treasury) and UK (DMO) described in the companion reference documents, Eurozone fiscal architecture remains fundamentally decentralised — each member state issues its own sovereign debt under its own fiscal framework, subject to EU-level fiscal rules (the Stability and Growth Pact, as reformed) rather than a single unified fiscal authority.
- **NextGenerationEU and joint EU issuance:** the EU's post-pandemic recovery fund, funded through joint EU-level bond issuance, represents a structurally novel (though still comparatively limited in scale relative to aggregate member-state debt) development in EU-level joint fiscal capacity, watched by analysts as a potential, gradual step toward greater fiscal integration that could, over time, affect the term premium and fragmentation-risk dynamics described in Sections 3.2 and 8.1.

10. FEI vs SR3 vs SO3 — A Three-Way Structural Comparison

Because Euribor, SOFR, and SONIA futures together constitute the three major Western STIR futures markets covered across this document series, this section catalogues the specific structural differences a trader moving between any pair of these markets should hold in mind, presented as a factual comparison only.

Dimension	FEI (Eurozone EURIBOR)	SR3 (US SOFR)	SO3 (UK SONIA)
Underlying rate type	Forward-looking term rate (panel-submitted, hybrid methodology)	Backward-looking compounded overnight secured rate	Backward-looking compounded overnight unsecured rate
IBOR transition outcome	Reformed, not discontinued — survives alongside €STR	Discontinued — fully replaced by SOFR	Discontinued — fully replaced by SONIA
Settlement mechanism	Single point-in-time fixing on last trading day	Compounded daily rate over a 3-month accrual period	Compounded daily rate over a 3-month accrual period
"Days in quarter" attribution needed?	No — direct forward-curve expression	Yes — central to spread/fly interpretation	Yes — central to spread/fly interpretation
Central bank / policy committee	ECB Governing Council	Federal Reserve / FOMC	Bank of England / MPC
Number of economies governed	20 member states, one currency	1 (US)	1 (UK)
Exchange(s)	ICE Futures Europe (FEI) and Eurex (FEU3) — dual-listed	CME Group (single venue)	ICE Futures Europe (single venue)
Risk-free overnight rate	€STR — separate benchmark, separate futures contract	SOFR — the IBOR replacement itself	SONIA — the IBOR replacement itself
Fragmentation-risk dimension	Yes — sovereign spread monitoring (BTP-Bund etc.) unique to currency union	No direct equivalent	No direct equivalent

10.1 Why the Differences Matter for Cross-Market Positioning

- A trader running a relative-value position across FEI and either SR3 or SO3 is, by construction, also taking a EUR/USD or EUR/GBP view, the same general principle described for the SO3-SR3 comparison in the companion document, here extended to a three-way comparison.
- The settlement-mechanism difference (single fixing for FEI versus compounded accrual for SR3/SO3) means a basis trade between FEI and either SOFR or SONIA futures requires careful attention to how each instrument's sensitivity to a given central bank decision date differs mechanically, not merely to the rate-level differential between the two currencies.
- The currency union's fragmentation-risk dimension means FEI positioning, at the margin, can be affected by idiosyncratic member-state political or fiscal developments (an Italian budget dispute, for example) with no analogue in the single-country SR3 or SO3 frameworks, a genuinely distinct source of tail risk specific to the Eurozone STIR market.

11. Summary Data Release Calendar

Release	Publisher	Frequency	Typical Timing	Primary FEI Curve Impact
HICP Flash Estimate	Eurostat	Monthly	Last business day of month, ~11:00 CET	All curve — largest single monthly event
ECB Governing Council Decision + Press Conference	ECB	8× per year	14:15 / 14:45 CET	All curve — immediate repricing; press conference often the larger move
ECB Staff/Eurosystem Projections	ECB	4× per year, with Governing Council decision	With Mar/Jun/Sep/Dec decisions	C2-C12 — growth/inflation forecast revision
National flash HICP (DE, FR, ES, IT)	National statistical offices	Monthly, staggered	Days before Eurozone flash	C1-C4 — early read on the aggregate figure
S&P; Global Eurozone Composite PMI	S&P; Global	Monthly	~3rd-5th business day	C1-C4 — activity leading indicator
German Ifo Business Climate	Ifo Institute	Monthly	~Last week of month	C1-C4 — German manufacturing/export sentiment
Eurozone GDP (flash + final)	Eurostat	Quarterly	~Mid-month following quarter end	C2-C8 — growth trajectory with national breakdowns
ECB Negotiated Wage Tracker	ECB	Quarterly	Variable	C2-C8 — forward-looking wage pressure signal
ECB Account of the Monetary Policy Meeting	ECB	8× per year, ~4-week lag	Variable	C3-C8 — debate nuance
ECB Bank Lending Survey	ECB	Quarterly	~3 weeks before Governing Council meeting	C2-C8 — credit conditions, transmission strength
BTP-Bund and peripheral sovereign spreads	Market, continuous	Continuous	Market hours	All curve, acutely during stress — fragmentation-risk signal
EMMI daily EURIBOR fixing	European Money Markets Institute	Daily	~11:00 CET	Front contract — daily fixing reference
€STR daily publication	ECB	Daily	~09:00 CET	Cross-market — €STR/EURIBOR spread monitoring

Three-Month Euribor futures occupy a genuinely distinct structural position among the major Western STIR futures markets covered in this document series. Where SR3 and SO3 both emerged from a clean IBOR discontinuation and replacement, Euribor survived its own reform process as a live, forward-looking, panel-submitted term rate, coexisting alongside — rather than being replaced by — the Eurozone's own risk-free overnight benchmark, €STR. This gives the FEI curve a simpler settlement mechanic (no accrual-period "days in quarter" arithmetic) but layers on top of it the unique governance complexity of a single monetary policy spanning twenty member states, with sovereign fragmentation risk as a structural consideration entirely absent from the single-country US and UK frameworks.

For spread traders, the most reliable framework treats FEI as governed by the same general ECB-meeting-driven repricing logic described for SR3 and SO3 — track the HICP flash and Governing Council calendar as the primary triggers, use the simpler direct-forward-curve spread interpretation enabled by the single-fixing settlement mechanism, and monitor the EURIBOR-€STR spread as an interbank credit/liquidity stress signal — while remaining alert to the currency union's structural fragmentation-risk dimension (BTP-Bund and other peripheral spreads) as a source of tail risk and Governing Council policy constraint with no parallel in either the SR3 or SO3 markets.